ATHILA ASHRAF

Full Stack Developer | AI Applications

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Visa Status: UAE Dependent Visa

PROFESSIONAL SUMMARY

MCA graduate (2026) who builds, deploys and maintains Python web applications with AI features. Delivered a paid stock and billing system to a wholesale client, deployed a live LLM tutoring app on Render, and fixed a production failure caused by a deprecated model. Built a real-time traffic monitoring system (89.3% mAP@0.5) and presented the related research at an international conference. Works with Flask, Django, REST APIs, SQL and Git, and keeps secrets out of code with environment variables.

TECHNICAL SKILLS

Back End & APIs: Python, Flask, Django, REST APIs, third-party API integration, authentication (admin login, hashed credentials), SQLAlchemy.
AI: LLM integration (Groq APIs), Scikit-learn, YOLOv8, OpenCV.
Front End: HTML5, CSS3, JavaScript, Bootstrap.
Databases: SQL (MySQL, SQLite).
Deployment & Maintenance: Render, Gunicorn, environment variables, debugging and fixing production issues, Git, GitHub, Linux.
Security: Secure development practices, secrets management, TCP/IP, DNS, HTTP/HTTPS

FREELANCE EXPERIENCE

Freelance Web Developer – Wholesale Clothing Brand | Sep 2026 – Oct 2026
Technologies: Python, Flask, SQLAlchemy, SQLite, HTML, CSS, Git

GitHub Repository

- Automated billing with percentage or fixed discounts, PDF invoice generation and bill cancellation that restores stock automatically.
- Implemented low-stock alerts and logged every stock correction for traceability.
- Gathered requirements from the client, then designed, built and delivered the stock management and billing application, installed on the client's PC.
- Secured the dashboard behind an admin login, with secrets kept in environment variables.

PROJECT EXPERIENCE

Smart Traffic Monitoring System Using Deep Learning – MCA Final Year Project | 2026
Technologies: Python, Flask, YOLOv8, SORT, OpenCV, NumPy, HTML, CSS, JavaScript, Git

GitHub Repository

- Built a real-time vehicle detection and tracking system with YOLOv8 and SORT, achieving 89.3% mAP@0.5, 88.6% precision and 86.4% recall at 42 ms inference per frame.
- Achieved 92.8% lane-wise vehicle counting accuracy across six test videos, applying PCU-weighted traffic density for congestion analysis.
- Developed a Flask backend with role-based admin and police dashboards, traffic trend prediction, adaptive signal timing simulation and automated email alerts (smtplib).

LEARNIQ – AI-Powered Adaptive Tutoring System – Personal Project | 2026
Technologies: Python, Flask, Groq LLaMA 3.1 8B, HTML, CSS, JavaScript, python-dotenv

GitHub Repository | **Live Demo**

- Built and deployed a live Flask application (Render + Gunicorn) with an HTML, CSS and JavaScript front end, integrating the Groq API to generate explanations, quizzes and feedback.
- Traced a live outage to a discontinued Groq model, updated the model configuration and redeployed on Render, restoring the app without changing its features.
- Engineered a 3-prompt pipeline (explain, question, evaluate) with low-temperature inference (0.4) for consistent output across subjects and grade levels.
- Secured API keys and secrets in environment variables, with none in the repository.

VOXEN – Voice-Controlled Local AI Agent – Internship Assignment Project | 2026
Technologies: Python, Streamlit, Groq Whisper Large V3, Groq LLaMA 3.3 70B

GitHub Repository | **Demo Video**

- Built an end-to-end voice agent: live microphone or uploaded audio is transcribed with Groq Whisper Large V3, classified by LLaMA 3.3 70B into four intents (code generation, file creation, summarization, chat), and executed by a modular tool engine.
- Evaluated local Whisper on CPU, found it too slow and memory-heavy for real-time use on an 8 GB machine, and switched to a hosted speech-to-text API, documenting the trade-off in the README.
- Separated speech-to-text, intent detection and tool execution into independent modules, with a Streamlit UI showing the transcript, intent and result of each request.

SymptoCare – Hospital Management System with AI Disease Prediction | Sep 2025 – Dec 2025
Technologies: Python, Django, MySQL, Machine Learning, HTML, CSS, JavaScript

GitHub Repository

- Built a full-stack Django and MySQL hospital management system with a normalized 13+ table database and REST APIs.
- Implemented role-based modules for Admin, Doctor, Hospital and Patient users.
- Trained a Random Forest model on a 4,920-row dataset (132 symptom features, 41 diseases) and integrated it into the Django application for end-to-end disease prediction.

RESEARCH & CONFERENCE PRESENTATIONS

An Analytical Study of Deep Learning Techniques in Road Traffic Analysis
Presented at the two-day International Conference Synergy 2026, organized by MTM College of Arts, Science and Commerce in collaboration with Sreenarayanaguru Open University. February 2026

EDUCATION

Master of Computer Applications (MCA) **2026**
MES College of Engineering, Kuttippuram

- CGPA: 8.15

Bachelor of Computer Applications (BCA) **2024**
University of Calicut

- CGPA: 8.025

COURSES & CERTIFICATIONS

- Google Cybersecurity Professional Certificate – Google, via Coursera (In Progress)
- Cloud Computing – NPTEL, IIT Kharagpur (2026)
- Essentials of Git – Sync N Spark (2024)
- Artificial Intelligence Workshop – Hands-on Training (2022)